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Valid Linear Inequalities for Fixed Charge Problems

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Many problems in the Operations Research/Management Science literature can be formulated with both zero-one and continuous variables. However, the exact optimization of such mixed zero-one models remains a computational challenge. In this paper, we propose to study mixed problems from a mathematical point of view that is similar in spirit to recent research on purely combinatorial problems that has investigated systems of defining linear inequalities (or facets of the underlying polytope). At least two numerical studies have validated this line of research computationally, and the advances in the problem-solving capabilities are considerable. We expect that similar gains are possible for the mixed zero-one problem. More precisely, we consider the mixed integer programs whose feasible region X is composed of (i) a simple additive constraint in the continuous variables x_j for $j = 1, 2, \cdots, n$ and (ii) constraints $0 \leq x_j \leq m_j y_j$ defined by binary variables y_j for $j = 1, 2, \cdots, n$. This type of feasible region arises in a variety of mixed integer problems, and particularly in network problems with fixed charges on the arcs. We derive two classes of facet-defining linear inequalities of the convex hull of X, and show that the second of these classes gives a complete description of the convex hull when $m_j = m$ for all j. We also develop methods to detect violated inequalities from these classes, so that these facets can be used as cutting planes to strengthen the formulations of certain mixed integer problems.

TEXTBOOKS IN Operations Research/Management Science generally describe three methods for resolving integer and mixed-integer optimization problems: branch-and-bound, cutting plane methods, and implicit or partial enumeration. Of these three methods only branch-and-bound has, over the past 20 years, been incorporated in commercial codes, and despite the occasional success of a purely branch-and-bound based solution procedure, it is widely accepted that branch-and bound alone is an insufficient tool for resolving real-world, large-scale integer programming problems. Indeed, most practitioners of branch-and-bound

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readily admit that branch-and-bound can in general be sped up considerably by the addition of suitably selected cutting planes and frequently, as well, by a reformulation of the problem to be solved. For instance, it appears now to be accepted that it is worthwhile to add to a constraint such as

$$12x_1 + 15x_2 + 16x_3 + 5x_4 + 34x_5 \leq 50, \quad x_i \in \{0, 1\}, \quad i = 1, \cdots, 5$$

the valid inequalities

$$x_1 + x_2 + x_5 \leq 2, \quad x_1 + x_3 + x_5 \leq 2, \quad x_2 + x_3 + x_5 \leq 2,$$
$$x_1 + x_2 + x_3 + 2x_5 \leq 3,$$

and that a constraint such as

$$49x_1 + 12x_2 + 33x_3 + 9x_4 + 11x_5 \geq 30, \quad x_i \in \{0, 1\}, \quad i = 1, \cdots, 5$$

should be replaced by the equivalent constraint having smaller coefficients

$$30x_1 + 12x_2 + 30x_3 + 9x_4 + 11x_5 \geq 30, \quad x_i \in \{0, 1\}, \quad i = 1, \cdots, 5$$

or, even better,

$$3x_1 + x_2 + 3x_3 + x_4 + x_5 \geq 3, \quad x_i \in \{0, 1\}, \quad i = 1, \cdots, 5$$

if a method related to linear programming (such as branch-and-bound) is used for resolution.

The problem of reformulating a single inequality in 0-1 variables and replacing it by an equivalent inequality with smaller coefficients has been examined in Bradley et al. [1974]. The problem of finding suitable valid constraints that tighten the problem formulation has been treated by a number of authors over the past 10 years: see Balas [1975], Hammer et al. [1975], Padberg [1980], Wolsey [1975, 1976], Zemel [1978], and Padberg [1979] for a related survey and references. In short, these studies are concerned with finding a (partial) characterization of the convex hull of the 0-1 solutions by way of linear inequalities.

There are some interesting ad hoc computational results on reformulation and problem "tightening" reported in Williams [1974, 1978] which were motivated to some extent by the earlier theoretical studies. However, the major computational validation for these theoretical developments has appeared only recently (Crowder et al. [1983]). This computational study carried out coefficient reduction automatically and generated valid inequalities (or facets), such as those shown in the examples above, in a systematic way, permitting the solution to optimality of a series of 0-1 problems that could not be solved otherwise by a commercial state-of-the-art branch-and-bound code.

The aim in this paper is to begin to tackle mixed 0-1 problems in a

similar fashion by studying linear inequalities (facets) which are necessary for a linear description of the convex hull of the solutions that consist of both 0-1 and real variables. For instance, the replacement of the constraints

$$x_1 + x_2 + x_3 \leq 3y; \quad 0 \leq x_i \leq 1; \quad i = 1, 2, 3; \quad y \in \{0, 1\}$$

by the constraints

$$0 \leq x_i \leq y; \quad i = 1, 2, 3; \quad y \in \{0, 1\},$$

which is now standard for location problems, is a simple example of what we have in mind.

Fixed charge and concave cost network problems have been a major motivation for this paper. Branch-and-bound methods have been notoriously unsuccessful on such problems, so our initial aim is to improve the formulation of such problems. A first attempt in this direction can be found in de Groote, and the well-known fact that the basic arcs of fixed charge networks form a spanning tree is one reason for the examination of the tree polytope in Padberg and Wolsey [1984].

Here, we propose to study three sets:

$$X_{\leq} = \{(x, y) \in R^{2n} \mid \textstyle\sum_{j=1}^{n} x_j \leq d,\ 0 \leq x_j \leq m_j y_j,\ y_j \in \{0, 1\}, j = 1, \cdots, n\}$$

$$X_{=} = \{(x, y) \in R^{2n} \mid \textstyle\sum_{j=1}^{n} x_j = d,\ 0 \leq x_j \leq m_j y_j,\ y_j \in \{0, 1\}, j = 1, \cdots, n\}$$

and

$$X_{\geq} = \{(x, y) \in R^{2n} \mid \textstyle\sum_{j=1}^{n} x_j \geq d,\ 0 \leq x_j \leq m_j y_j,\ y_j \in \{0, 1\}, j = 1, \cdots, n\}$$

and their associated convex hulls, $\mathscr{P}_{\leq} = \text{conv}(X_{\leq})$, $\mathscr{P}_{=} = \text{conv}(X_{=})$, and $\mathscr{P}_{\geq} = \text{conv}(X_{\geq})$.

Such sets (or subsets of such sets) arise typically in fixed cost network problems when one looks at a single node. Arcs with flow x_j and capacity m_j for $j = 1, \cdots, n$ lead into the node, and a quantity d (or $\leq d$, or $\geq d$) must flow out.

The sets $X_{\cdot}$ are also found in a variety of other models. In capacitated location problems, x_j for $j = 1, \cdots, n$ may be the flow out of a warehouse with stock m_j, while d is the aggregate demand of the customers. In production problems, x_j for $j = 1, \cdots, n$ may denote the amount produced on machine j whose capacity is m_j, while d is the total demand in some period.

In Section 2, we derive a class of "surrogate knapsack" facets for $\mathscr{P}_{\geq}$. The polytope $K = \text{conv}\{y \in R^n\colon \sum_{j=1}^{n} m_j y_j \geq d, y_j \in \{0, 1\}\ j = 1, \cdots, n\}$ is a relaxation of $\mathscr{P}_{\geq}$ and $\mathscr{P}_{=}$, and we show that "almost all" facets of K are facets of $\mathscr{P}_{\geq}$.

In Section 3, we describe a class of mixed integer facets for $\mathscr{P}_=$ called "flow cover" facets. These facets are of the form:

$$\sum_{j\in S} x_j - \sum_{j\in S} \mu_j y_j \leq v$$

with $\sum_{j\in S} m_j > d$, and $S \subseteq N = \{1, \cdots, n\}$.

In Section 4, we examine some general properties of the facets of $\mathscr{P}_\leq$, $\mathscr{P}_=$ and $\mathscr{P}_\geq$. In particular, we show how to construct facets of $\mathscr{P}_\leq$ and $\mathscr{P}_\geq$ from facets of $\mathscr{P}_=$.

Clearly, when the set $X_.$ is only part (a relaxation) of a larger model, facets of $\mathscr{P}_.$ are not necessarily facets for the larger model. However, our experience suggests that it is crucially important to use facets of subproblems whenever they are available.

The problem of using "surrogate knapsack" inequalities as cutting planes has essentially been treated in Crowder et al. for $\mathscr{P}_\geq$ and $\mathscr{P}_=$. In Section 5, we describe how "surrogate knapsack" inequalities can be generated for $\mathscr{P}_\leq$, and how violated "flow cover" facets can be found.

Finally, in Section 6, we show that when $m_j = m$ for all $j = 1, \cdots, n$, the class of "flow cover" facets gives a complete description of the convex hull of $\mathscr{P}_=$, and we terminate with a brief discussion of extensions, and an example.

To rule out uninteresting special cases, we make the blanket assumptions

$$\left.\begin{aligned} \textstyle\sum_{j=1}^n m_j &> d + m_k \\ 0 < m_k &\leq d \end{aligned}\right\} \quad \text{for all} \quad k \in N, \qquad \begin{aligned}&\text{(A1)}\\&\text{(A2)}\end{aligned}$$

throughout the paper. Consequently, dim $\mathscr{P}_\leq$ = dim $\mathscr{P}_\geq$ = $2n$, and dim $\mathscr{P}_= = 2n - 1$. This result, in turn, implies that both $\mathscr{P}_\leq$ and $\mathscr{P}_\geq$ have a *unique* representation with facet-defining linear inequalities, whereas the representation of $\mathscr{P}_=$ is unique modulo multiples of the constraint $\sum_{j=1}^n x_j = d$.

The defining inequalities $x_j \geq 0$, $x_j \leq m_j y_j$ and $y_j \leq 1$ for $j \in N$ are the *trivial* facets of the three polytopes. The proof of this statement uses the conditions (A1) and (A2), but is standard and therefore omitted. Furthermore, the inequality $\sum_{j\in N} x_j \leq d$ ($\sum_{j\in N} x_j \geq d$, respectively) is a facet of $\mathscr{P}_\leq$ (of $\mathscr{P}_\geq$, respectively) which we will also call trivial. Finally, the equation $\sum_{j\in N} x_j = d$ defines the smallest affine subspace of R^{2n} that contains $\mathscr{P}_=$.

1. THE SURROGATE KNAPSACK PROBLEM AND FACETS FOR $\mathscr{P}_\geq$

The first set of valid inequalities we study involves only the 0-1 variables y_j for $j \in N$. We consider the "surrogate knapsack" problem:

$$\sum_{j=1}^n m_j y_j \geq d, \quad y_j \in \{0, 1\}, \quad j \in N,$$

and the associated knapsack polytope:

$$K = \text{conv}\{y \in R^n \mid \textstyle\sum_{j=1}^n m_j y_j \geq d,\ y_j \in \{0, 1\},\ j \in N\}.$$

PROPOSITION 1. *Every valid inequality for K is a valid inequality for $\mathscr{P}_{\geq}$.*

Proof. Suppose that the point $(x, y) \in X_{\geq}$. Then $\sum_{j=1}^n x_j \geq d$ and $x_j \leq m_j y_j$. Therefore, $\sum_{j=1}^n m_j y_j \geq d$, and $y_j \in \{0, 1\}$ for $j \in N$. Hence, $y \in K$.

The obvious question now is to ask when facets of K define facets of $\mathscr{P}_{\geq}$.

THEOREM 2. *Let $\mu y \geq 1$ be a facet of K. Then $\mu y \geq 1$ is a nontrivial facet of $\mathscr{P}_{\geq}$ if and only if, for every $j \in N$, there exists some point $(x^j, y^j) \in X_{\geq}$ with $\mu y^j = 1$, $y_j{}^j = 1$, and $x_j{}^j < m_j$.*

Proof. We first show necessity. Suppose $\mu y \geq 1$ is a nontrivial facet of $\mathscr{P}_{\geq}$, but the condition of the theorem is violated. Then for some $j \in N$, all points $(x, y) \in X_{\geq}$ satisfying $\mu y = 1$ have $x_j = m_j y_j$. Since this equality is a trivial facet, and the facets are uniquely defined, we have a contradiction.

To prove sufficiency, we aim to find n points $(\tilde{x}^i, \tilde{y}^i) \in X_{\geq}$ such that $\{\tilde{y}^i\}_{i=1}^n$ define the given facet $\mu y \geq 1$ of K, and the n points satisfy

CONDITION A. *For each j there exists $i(j) \in N$ so that $\tilde{x}_j^{i(j)} < m_j \tilde{y}_j^{i(j)}$.*

The result will then follow, because for positive ε sufficiently small, the $2n$ points $(\tilde{x}^i, \tilde{y}^i)$ for $i = 1, \cdots, n$ and $(\tilde{x}^{i(j)} + \varepsilon e_j, \tilde{y}^{i(j)})$ are linearly independent, are in $X_{\geq}$, and satisfy $\mu y = 1$.

Hence, it remains to show that if the condition of the theorem is satisfied, we can find n linearly independent points $(\hat{x}^i, \hat{y}^i)$ satisfying Condition A. We do so by an iterative procedure.

We start with n zero-one points y^i for $i = 1, \cdots, n$ defining the facet $\mu y \geq 1$ of K. Define $\{x^i\}_{i=1}^n$ as follows:

$$x_j^i = d m_j y_j^i / \textstyle\sum_{k=1}^n m_k y_k^i, \quad j = 1, \cdots, n.$$

Since $y^i \in K$, $\sum_{k=1}^n m_k y_k^i \geq d$, so that $x_j^i \leq m_j y_j^i$ for all i and j, and the n points (x^i, y^i) for $i = 1, \cdots, n$ lie in $X_{\geq}$.

Note that the n points satisfy the following property:

PROPERTY B. *If for some $j \in N$ there exists an index i satisfying $x_j^i = m_j$, then $x_j^i = m_j y_j^i$ for all j.*

This result follows from the above construction by observing that $y_j^i \in \{0, 1\}$ for all i and j and $\sum_{j=1}^n x_j^i = d$ for all i. We define next a sequence $\{(x^{il}, y^{il})_{i=1,\cdots,n}\}$ of points of $X_{\geq} \cap \{(x, y) \mid \mu y = 1\}$ so that y^{il} for $i = 1, \cdots, n$ are linearly independent, so that Property B remains

true, and so that for some $l \leq n$ Condition A is satisfied. We initialize $x^{i0} = x^i$, $y^{i0} = y^i$ for $i = 1, \cdots, n$. Suppose now that for $l \geq 0$, the points (x^{il}, y^{il}) for $i = 1, \cdots, n$ satisfy Property B and are such that y^{il} for $i = 1, \cdots, n$ are linearly independent. Define

$$K^l = \{j \in N \mid \exists i(j) \text{ such that } x_j^{i(j)l} < m_j y^{i(j)l}\}.$$

If $K^l = N$, Condition A is satisfied and we stop. Otherwise, let $j \in N - K^l$. Then, by Property B we have that for all i such that $y_j^{il} = 1$, $x_k^{il} = m_k y_k^{il}$ for all $k \in N$. Let $H_j^l = \{i \in N \mid y_j^{il} = 1\}$. Clearly, $H_j^l \neq \phi$, because the n points y^{il} are linearly independent. By the assumption of the theorem, there exists a point $(x^*, y^*) \in X_{\geq} \cap \{(x, y) \mid \mu y = 1\}$ satisfying $y_j^* = 1$ and $0 \leq x_j^* < m_j$. Then, there exists $\lambda \neq 0$ such that $y^* = \sum_{i=1}^n \lambda_i y^{il}$. Since $y_j^* = 1$, we have $\sum_{i \in H_j^l} \lambda_i = 1$ and thus there exists $i_* \in H_j^l$ with $\lambda_{i_*} = 0$. Standard linear algebra arguments imply that $y^{1l}, \cdots, y^{i_*-1,e}, y^*, y^{i_*+1,e}, \cdots, y^{nl}$ are linearly independent. Define $(\hat{x}, \hat{y})$ as

$$\hat{x}_j = x_j^* + \varepsilon < m_j, \quad \hat{x}_k = x_k^* - \varepsilon/n \quad \text{whenever} \quad x_k^* = m_k,$$

$$\hat{x}_k = x_k^* \quad \text{otherwise}, \quad \hat{y} = y^*,$$

where $\varepsilon > 0$. It is clear that $(\hat{x}, \hat{y}) \in X_{\geq} \cap \{(x, y) \mid \mu y = 1\}$ and that $\hat{x}_k < m_k$ for all k. We define $x^{il+1} = x^{il}$, $y^{il+1} = y^{il}$ for all $i \neq i_*$ and $x^{i_*l+1} = \hat{x}$, $y^{i_*l+1} = \hat{y}$. Then we have n linearly independent points contained in $X_{\geq} \cap \{(x, y) \mid \mu y = 1\}$. We claim that $(x^{i,l+1}, y^{i,l+1})$ for $i = 1, \cdots, n$ have the Property B. Suppose not. Then there exist some $j \in N$ and some $i \in N$ with $x_j^{i,l+1} = m_j$ such that $x_k^{i,l+1} < m_k y_k^{i,l+1}$ for at least some $k \neq j$. Since row i_* satisfies $x_k^{i_*,l+1} < m_k$ for all k, we have $i \neq i_*$. But then (x^{il}, y^{il}) for $i = 1, \cdots, n$ did not have Property B, a contradiction. Thus, we can repeat the argument. Since

$$K^{l+1} = K^l \cup \{j \in N \mid 0 < \hat{x}_j\}$$

satisfies $|K^{l+1}| > |K^l|$, we stop after at most n iterations, with points (x^i, y^i) for $i = 1, \cdots, n$ satisfying all the desired properties. This conclusion completes the proof of the theorem.

2. THE CLASS OF "FLOW COVER" FACETS FOR $\mathscr{P}_=$

In contrast to the inequalities of the previous section, those derived in this section involve both the continuous and integer variables. We say that $S \subseteq N$ is a *cover* if $\sum_{j \in S} m_j > d$, and we associate the value $\lambda = \sum_{j \in S} m_j - d$ with each cover.

We first describe a large class of valid inequalities called "flow cover" inequalities for $\mathscr{P}_{\leq}$.

PROPOSITION 3. *Let S be a cover with $\bar{m} = \max_{j \in S} m_j$ and $L \subseteq N \backslash S$. Let $\bar{m}_j = \max(\bar{m}, m_j)$ and let $(x)^+$ denote* $\max(x, 0)$. *Then the (S, L) inequality*

$$\sum_{j \in S \cup L} x_j + \sum_{j \in S} (m_j - \lambda)^+ (1 - y_j) \leq d + \sum_{j \in L} (\bar{m}_j - \lambda) y_j$$

is a valid inequality for $\mathscr{P}_{\leq}$.

Proof. Let $S^+ = \{j \in S: m_j > \lambda\}$, and let (x, y) be a point of $X_{\leq}$ with $y_i = 1$ for $i \in T$, and $y_i = 0$ otherwise.

Case 1. $|S^+ \backslash T| \leq |L \cap T|$.

$$\begin{aligned} &\sum_{j \in S \cup L} x_j + \sum_{j \in S} (m_j - \lambda)^+ (1 - y_j) - \sum_{j \in L} (\bar{m}_j - \lambda) y_j \\ &\qquad \leq d + |S^+ \backslash T| (\bar{m} - \lambda) - |L \cap T| (\bar{m} - \lambda) \leq d \end{aligned}$$

using first $\sum_{j \in S \cup L} x_j \leq \sum_{j \in N} x_j \leq d$, $m_j \leq \bar{m}$ for $j \in S^+$, and $-\bar{m}_j \leq -\bar{m}$ for $j \in L$, and then $\bar{m} \geq \lambda$ and $|S^+ \backslash T| \leq |L \cap T|$.

Case 2. $|S^+ \backslash T| \geq |L \cap T| + 1$.

$$\begin{aligned} &\sum_{j \in S \cup L} x_j + \sum_{j \in S} (m_j - \lambda)^+ (1 - y_j) - \sum_{j \in L} (\bar{m}_j - \lambda) y_j \\ &\qquad \leq \sum_{j \in S \cap T} m_j + \sum_{j \in L \cap T} m_j + \sum_{j \in S^+ \backslash T} m_j - |S^+ \backslash T| \lambda \\ &\qquad\quad - \sum_{j \in L \cap T} \bar{m}_j + |L \cap T| \lambda \\ &\qquad \leq \sum_{j \in S} m_j + \sum_{j \in L \cap T} (m_j - \bar{m}_j) - \lambda \\ &\qquad\quad + \lambda \{|L \cap T| + 1 - |S^+ \backslash T|\} \\ &\qquad \leq d \end{aligned}$$

using first $x_j \leq m_j$ for $j \in S \cup L$, then $(S \cap T) \cap (S^+ \backslash T) \subseteq S$, and finally $\sum_{j \in S} m_j - \lambda = d$, $m_j - \bar{m}_j \leq 0$ for $j \in L \cap T$, and $|L \cap T| + 1 - |S^+ \backslash T| \leq 0$.

At this stage, is it perhaps worth noting that the (S, L) inequality is of interest only for those $j \in L$ for which $m_j > \bar{m} - \lambda$, as otherwise the inequality obtained by combining the flow cover inequality with $S' = S$, $L' = L - \{j\}$, and the inequality $x_j \leq m_j y_j$, dominates the (S, L) inequality.

We now consider situations in which the above inequalities are facets for $\mathscr{P}_{=}$. For simplicity, we give the result in two steps, and treat first the case $L = \phi$.

THEOREM 4. *Let S be a cover, with $\bar{m} = \max_{j \in S} m_j > \lambda$, then*

$$\sum_{j \in S} x_j + \sum_{j \in S} (m_j - \lambda)^+ (1 - y_j) \leq d$$

is a facet of $\mathscr{P}_{=}$.

Proof. As $\sum_{j \in S} m_j > d$, there exist $s = |S|$ linearly independent points $X^1, \cdots, X^s$ satisfying $0 \leq x_j \leq m_j$ for $j \in S$, and $\sum_{j \in S} x_j = d$.

Now let $\xi_k = \max\{\sum_{j\in S} x_j + \sum_{j\in S\setminus k} (m_j - \lambda)^+(1 - y_j) \mid \sum_{j\in S} x_j \le d,\ x_j \le m_j y_j$ for $j \in S$, $y_j \in \{0, 1\}$ for $j \in S$, $y_k = 0\}$, and let $\tilde{X}^k$ be an optimal solution for the x variables. We claim that $\xi_k = d - (m_k - \lambda)^+$. Using Proposition 3 with $L = \phi$ and $y_k = 0$ gives $\xi_k + (m_k - \lambda)^+ \le d$. However, setting $y_j = 1$ for all $j \in S\setminus k$, we see that $\xi_k \ge \min\{d, \sum_{j\in S} m_j - m_k\} = \min\{d, d + \lambda - m_k\} = d - (m_k - \lambda)^+$, and the claim is justified.

Now from Assumption A1, $\sum_{j=1}^n m_j - m_k > d$ for all $k \in N$. Combining this result with the assumption that $\max_{j\in S} m_j > \lambda$, we see that $S \subsetneqq N$. We suppose that $m_1 > \lambda$, $1 \in S = \{1, \cdots, s\}$. For each vector $\tilde{X}^k$ with $\sum_{j\in S} \tilde{X}_j^k = d - (m_k - \lambda)^+$, define $\varepsilon_k \ge 0$ such that $\sum_{j\in S} \tilde{X}_j^k + \varepsilon_k \sum_{j\in N\setminus S} m_j = d$. Let $\bar{X}^k = \varepsilon_k(m_{s+1}, \cdots, m_n)$. Note that $\varepsilon_1 > 0$.

Let e_j denote the jth unit vector and let $\mathbf{1}$ denote a vector whose components are all one. Now the following set of $2n - 1$ points

$$
\begin{array}{ll}
(X^k, \mathbf{1}, 0, 0) & k = 1, \cdots, s \\
(\tilde{X}^k, \mathbf{1} - e_k, \bar{X}^k, \mathbf{1}) & k = 1, \cdots, s \\
(X^1, \mathbf{1}, 0, e_j) & j = s + 1, \cdots, n \\
(\tilde{X}^1, \mathbf{1} - e_1, \bar{X}^1 + \varepsilon(e_j - e_{j+1}), \mathbf{1}) & j = s + 1, \cdots, n - 1
\end{array}
$$

lies in $X_=$, and satisfies the inequality with equality.

We now suppose that these points lie on the hyperplane $\sum_{j\in N} \alpha_j x_j + \sum_{j\in N} \beta_j y_j = \alpha_0$. Since $(X^k, \mathbf{1}, 0, 0)$ for $k = 1, \cdots, s$ lie in the hyperplane, $\sum_{j\in S} \alpha_j X_j^k + \sum_{j\in S} \beta_j = \alpha_0$. These points uniquely define $\sum_{j\in S} x_j = d$, and hence we can take $\alpha_j = \alpha$ for $j \in S$, giving $\alpha d + \sum_{j\in S} \beta_j = \alpha_0$.

Consider now the points $(\tilde{X}^1, \mathbf{1}, 0, e_j)$ for $j = s + 1, \cdots, n$. Since $(X^1, \mathbf{1}, 0, 0)$ is tight, $\beta_j = 0$ for $j \in N\setminus S$.

Consider now the points $(\tilde{X}^1, \mathbf{1} - e_1, \bar{X}^1, \mathbf{1})$ and $(\tilde{X}^1, \mathbf{1} - e_1, \bar{X}^1 + \varepsilon(e_j - e_{j+1}), \mathbf{1})$ for $j = s + 1, \cdots, n - 1$. Then $0 = \alpha_j - \alpha_{j+1}$ for $j = s + 1, \cdots, n - 1$, or $\alpha_j = \theta$ a constant for $j \in N\setminus S$. Since we can arbitrarily add multiples of $\sum_{j\in N} x_j = d$, we take $\theta = 0$ without loss of generality.

Finally, we consider the points $(\tilde{X}^k, \mathbf{1} - e_k, \bar{X}^k, \mathbf{1})$. Substituting, we obtain: $\alpha \sum_{j\in S} \tilde{X}_j^k + \sum_{j\in S} \beta_j - \beta_k = \alpha_0$, or $\alpha(d - (m_k - \lambda)^+) + \sum_{j\in S} \beta_j - \beta_k = \alpha d + \sum_{j\in S} \beta_j$, or $\beta_k = -\alpha(m_k - \lambda)^+$. Hence, the $(2n - 1)$ points define the proposed inequality, which is a facet.

THEOREM 5. *The (S, L) inequality* (2) *is a facet of $\mathscr{P}_=$ if $0 < \bar{m} - \lambda < m_k \le \bar{m}$ for all $k \in L$.*

Proof. From Theorem 4 $\sum_{j\in S} x_j + \sum_{j\in S} (m_j - \lambda)^+(1 - y_j) \le d$ is a facet of $\mathscr{P}_= \cap \{(x_j, y_j) = (0, 0)$ for all $j \in L\}$. For each $k \in L$, we exhibit two points that lie on the inequality, and increase the rank of the set of linearly independent points by two. Again, we suppose that $\max_{j\in S} m_j = m_1 = \bar{m}$.

(i) Take $y_j = 1$ for $j \in S - \{1\} \cup k$, $y_j = 0$ otherwise; $x_k = m_k$, and $\sum_{j \in S-\{1\} \cup k} x_j = d$ with $x_j \leq m_j y_j$.
(ii) Take $y_j = 1$ for $j \in S - \{1\} \cup k$, $y_j = 0$ otherwise; $x_k = d - \sum_{j \in S-\{1\}} m_j$ and $\sum_{j \in S-\{1\} \cup k} x_j = d$, with $x_j \leq m_j y_j$.

From the assumptions, $\sum_{j \in S-\{1\}} m_j = d + \lambda - \bar{m} < d$. Also, $m_1 > \bar{m} - \lambda$ implies that $\sum_{j \in S-\{1\}} m_j + m_k > d$, so the points (i) and (ii) lie in $X_=$ and are distinct.

To see that the rank increases appropriately, it is sufficient to observe that for the matrix

$$U' = \left(\begin{array}{c|cc} U & \multicolumn{2}{c}{0} \\ \hline V & m_k & 1 \\ & x_k & 1 \end{array}\right)$$

with $0 \leq x_k < m_k$, rank $(U') \geq$ rank $(U) + 2$.

COROLLARY 6. *If $m_j = m$ for all j, we obtain an exponential number of facets for $\mathscr{P}_=$*

$$\sum_{j \in S \cup L} x_j + \sum_{j \in S} (m - \lambda)(1 - y_j) \leq d + \sum_{j \in L} (m - \lambda) y_j$$

where S runs through all minimum covers, and L all subsets of $N \backslash S$.

3. PROPERTIES RELATING $\mathscr{P}_\leq$, $\mathscr{P}_=$ AND $\mathscr{P}_\geq$

So far we have derived two classes of facets, the surrogate knapsack facets for $\mathscr{P}_\geq$, and the flow cover facets for $\mathscr{P}_=$. Evidently, the surrogate knapsack facets are all valid inequalities for $\mathscr{P}_=$, and as shown in Proposition 3 the flow cover facets are valid for $\mathscr{P}_\leq$. Furthermore, one might expect that for every facet of $\mathscr{P}_\geq$ there is a corresponding facet of $\mathscr{P}_\leq$, and vice versa. The results of this section give a partial answer to this question.

Below we shall use the following property:

PROPERTY (C1). *If $\pi x + \mu y \geq v$ is a nontrivial facet of $\mathscr{P}_r$, $r \in \{\leq, =, \geq\}$, then for each $j \in N$ there exists a point $(x, y) \in X_r$ lying on the facet with $x_j < m_j$ and $y_j = 1$ (i.e., $x_j < m_j y_j$).*

PROPOSITION 7. *If $\pi x - \mu y \leq v$ is a nontrivial facet of $\mathscr{P}_\leq$, then*
(i) $\pi, \mu \geq 0$, $v > 0$;
(ii) *If $\mu_j > 0$, then $\pi_j > 0$;*
(iii) $\sum_{j \in N} \{m_j \mid \pi_j > 0\} > d$.

Proof. (i) As the facet is nontrivial (not $y_j \leq 1$), there exists a point $(x, y) \in X_\leq$ on the facet with $y_j = 0$. The point $(x, y) + (0, e_j)$ is necessarily feasible, and hence $-\mu_j \leq 0$ for all $j \in N$. Similarly, there exists a point (x, y) with $x_j > 0$. Now $(x, y) - (\varepsilon e_j, 0)$ is feasible, and hence $\pi_j \geq 0$ for all j. Finally, since $(0, 0)$ is feasible, $v \geq 0$. If $v = 0$, the point $x_j = m_j$,

$y_j = 1$, and $x_k = y_k = 0$ for $k \neq j$ is feasible, and hence $\pi_j m_j - \mu_j \leq 0$. But then the nontrivial facet $\pi x - \mu y \leq 0$ is dominated by a linear combination of trivial facets, $\sum_{j=1}^{n} \pi_j(x_j - m_j y_j) \leq 0$, a contradiction.

(ii) Suppose $\pi_j = 0$ and $\mu_j > 0$. Let (x, y) be a point on the facet with $y_j = 1$. The point $(x, y) - (x_j e_j, e_j)$ is feasible, and so $-\pi_j x_j + \mu_j \leq 0$, a contradiction.

(iii) Let $S(\pi) = \{j \mid \pi_j > 0\}$. Restricted to the space of variables (x_i, y_i) for $i \in S(\pi)$, the inequality must still be a facet. Suppose that $\sum_{j \in S(\pi)} m_j \leq d$, and (x, y) is a point on the facet. Now (my, y) is feasible and since $\pi_j > 0$ for $j \in S(\pi)$, we must have $x_j = m_j y_j$ for all $j \in S(\pi)$, a contradiction.

PROPOSITION 8. *If $\pi x + \mu y \geq v$ is a nontrivial facet of $\mathscr{P}_{\geq}$, then*
(i) $\pi, \mu \geq 0$ *and* $v > 0$;
(ii) $\min_{j \in N} \pi_j = 0$;
(iii) $\sum_{j \in N} \{m_j \mid \pi_j = 0\} > d$.

The proofs, using Property C1, parallel closely those of Proposition 7.

PROPOSITION 9. *If $\pi x + \mu y \geq v$ is a nontrivial facet of $\mathscr{P}_{=}$, then $\mu \geq 0$.*

Propositions 7 through 9 give us an idea of the form of the nontrivial facets. Now we show how all the nontrivial facets of $\mathscr{P}_{=}$ give rise to facets of $\mathscr{P}_{\leq}$ and $\mathscr{P}_{\geq}$. Let $\mathbf{1}$ now denote an n-dimensional unit vector whose components are all ones.

PROPOSITION 10. *If $\pi x + \mu y \geq v$ is a nontrivial facet of $\mathscr{P}_{=}$, then*

$$(\pi - t\mathbf{1})x + \mu y \geq v - td$$

is a nontrivial facet of $\mathscr{P}_{\geq}$, where $t = \min\{\pi_j \mid j \in N\}$.

Proof. Let $(x, y) \in \mathscr{P}_{\geq}$. Then there exists $(\bar{x}, \bar{y}) \in \mathscr{P}_{=}$ such that $(\bar{x}, \bar{y}) \leq (x, y)$. Consequently, since $\mu \geq 0$ by Proposition 8(i), we have

$$\begin{aligned}(\pi - t\mathbf{1})x + \mu y &= \pi\bar{x} + \mu\bar{y} - td + (\pi - t\mathbf{1})(x - \bar{x}) + \mu(y - \bar{y}) \\ &\geq v - td\end{aligned}$$

and thus the inequality is valid for $\mathscr{P}_{\geq}$. Since $\pi x + \mu y \geq v$ is a facet of $\mathscr{P}_{=}$ there exist $2n - 1$ linearly independent points $(x^i, y^i) \in X_{=} \cap \{(x, y) \mid \pi x + \mu y = v\}$ where $i = 1, \cdots, 2n - 1$. Let $j \in N$ be such that $t = \pi_j$. By Property C1, choosing j such that $\pi_j = t$, there exists $(x, y) \in X_{=} \cap \{(x, y) \mid \pi x + \mu y = v\}$ satisfying $y_j = 1$ and $x_j < m_j$. Consequently, the point (x^*, y^*), defined to be the same as (x, y) except that $x_j^* = x_j + \varepsilon$ for small $\varepsilon > 0$, satisfies $(x^*, y^*) \in X_{\geq} \cap \{(x, y) \mid \pi x + \mu y = v\}$, $(x^*, y^*) \notin \mathscr{P}_{=}$. It follows that (x^*, y^*) is linearly independent of the $2n - 1$ points (x^i, y^i), where $i = 1, \cdots, 2n - 1$ and, furthermore, that the inequality is a nontrivial facet of $\mathscr{P}_{\geq}$.

Remembering that any inequality for $\mathscr{P}_=$ can be normalized by adding a multiple of $\sum_{j\in N} x_j = d$, we obtain:

PROPOSITION 11. *Let $\pi x + \mu y \geq v$ be a nontrivial facet of $\mathscr{P}_=$ normalized so that $\pi \leq 0$ and $\pi_j = 0$ for some $j \in N$. Then*

$$(\pi - t\mathbf{1})x + \mu y \geq v - td$$

is a nontrivial facet of $\mathscr{P}_\leq$, where $t = \max\{0, t_0\}$ and $t_0 = \max\{(v - \sum_{j=1}^n [\pi_j m_j + \mu_j] y_j)/(d - \sum_{j=1}^n m_j y_j) \mid \sum_{j=1}^n m_j y_j < d, y_j \in \{0, 1\}$ for $j \in N\}$.

Proof. We first prove that the proposed inequality is valid for $\mathscr{P}_\leq$. First, suppose $t = 0$, i.e., $t_0 \leq 0$. Then we have $\sum_{j=1}^n (\pi_j m_j + \mu_j) y_j \geq v$ for all $y_j \in \{0, 1\}$, such that $\sum_{j=1}^n m_j y_j < d$ and, consequently, $\pi x + \mu y \geq v$ for all $(x, y) \in \mathscr{P}_\leq$ satisfying $\sum_{j=1}^n m_j y_j < d$ because $\pi \leq 0$ holds. On the other hand, let $(x, y) \in \mathscr{P}_\leq$ be such that $\sum_{j=1}^n m_j y_j \geq d$. Then there exists $(\bar{x}, y) \in \mathscr{P}_=$ satisfying $\bar{x} \geq x$ and thus

$$\pi x + \mu y \geq \pi \bar{x} + \mu y \geq v,$$

and the assertion follows.

Now suppose $t > 0$. Then for $(x, y) \in \mathscr{P}_\leq$ with $\sum_{j=1}^n m_j y_j < d$, we have

$$\pi x + \mu y - t\mathbf{1}x \geq \sum_{j=1}^n (\pi_j m_j + \mu_j) y_j - t \sum_{j=1}^n m_j y_j \geq v - td$$

where the first inequality uses $\pi \leq 0$, and $x_j \leq m_j y_j$, and the second uses the definition of t. For the case $\sum_{j=1}^n m_j y_j \geq d$, let $(x, y) \in \mathscr{P}_\leq$ and $(\bar{x}, y) \in \mathscr{P}_=$ as above with $x \leq \bar{x}$. Then, we obtain

$$\pi x + \mu y - t\mathbf{1}x = \pi \bar{x} + \mu y - td + \pi(x - \bar{x}) + t\mathbf{1}(\bar{x} - x) \geq v - td$$

and thus validity of the proposed inequality for $\mathscr{P}_\leq$ follows.

To prove that the inequality defines a nontrivial facet, we first observe that there exist $2n - 1$ linearly independent points $(x^i, y^i) \in X_= \cap \{\pi x + \mu y = v\}$ where $i = 1, \cdots, 2n - 1$. Now if $t = 0$, we proceed as in the proof of Proposition 10, except that we choose $j \in N$ such that $\pi_j = 0$ and set $x_j^* = x_j - \varepsilon$, observing that there exists $(x, y) \in X_= \cap \{(x, y) \mid \pi x + \mu y = v\}$ such that $x_j > 0$. Suppose now that $t > 0$ holds. If $v - td = 0$, then the zero-vector satisfies the proposed inequality as an equality. Therefore $2n$ affinely independent points of $\mathscr{P}_\leq$ satisfy the inequality as equality, the inequality is thus a facet and hence by Proposition 7 a trivial facet of $\mathscr{P}_\leq$ different from $\sum_{j=1}^n x_j \leq d$. Because $\dim \mathscr{P}_\leq = 2n$, we have uniqueness of the facets of $\mathscr{P}_\leq$, and thus the inequality is one of the defining inequalities, a contradiction to $\pi \leq 0$ and $t > 0$. Consequently, $v - td \neq 0$. Now let y^* be any feasible zero-one point for which t_0 is attained and let $x_j^* = m_j y_j^*$ for $j \in N$. It follows that (x^*, y^*) satisfies the inequality as an equality. Because $(x^*, y^*) \notin X_=$ is linearly independent of the points (x^i, y^i) for $i = 1, \cdots, 2n - 1$, Proposition 11 follows.

Propositions 10 and 11 give a general construction of facets of $\mathscr{P}_{\leq}$ and $\mathscr{P}_{\geq}$ for each nontrivial facet of $\mathscr{P}_{=}$. In particular, looking at the flow cover facets for $\mathscr{P}_{=}$, we obtain

PROPOSITION 12. *Let S be a cover with $\bar{m} = \max_{j \in S} m_j > \lambda$, and $L \subseteq N \backslash S$ with $0 < \bar{m} - \lambda < m_k \leq \bar{m}$ for all $k \in L$,*

(i) $\sum_{j \in S \cup L} x_j + \sum_{j \in S} (m_j - \lambda)^+ (1 - y_j) \leq d + \sum_{j \in L} (\bar{m} - \lambda) y_j$ *is a facet of* $\mathscr{P}_{\leq}$

(ii) $\sum_{j \in N \backslash (S \cup L)} x_j + \sum_{j \in S} (m_j - \lambda)^+ y_j + \sum_{j \in L} (\bar{m} - \lambda) y_j \geq \sum_{j \in S} (m_j - \lambda)^+$ *is a facet of* $\mathscr{P}_{\geq}$.

Proof. Note that (i) is valid for $X_{\leq}$, so it suffices to find a tight point lying in $X_{\leq} \backslash X_{=}$. (ii) follows immediately from Proposition 10.

4. THE SEPARATION PROBLEM, OR GENERATING VIOLATED FACETS

Given a point (x^*, y^*), we wish to know whether $(x^*, y^*) \in \mathscr{P}_{\geq}$ ($\mathscr{P}_{=}$ or $\mathscr{P}_{\leq}$) or not. More specifically, we should like to find a surrogate knapsack, or a flow cover facet, cutting off the point (x^*, y^*).

The problem of testing whether (x^*, y^*) violates an $S - K$ facet for $\mathscr{P}_{\geq}$ is precisely the problem of testing whether y^* violates any facet of K. The basic result for this problem is

PROPOSITION 13 (Crowder et al.). *Let (η, z^T) be an optimal solution to the knapsack problem: $\eta = \min\{\sum_{j=1}^n y_j^* z_j : \sum_{j=1}^n m_j z_j > \sum_{j=1}^n m_j - d,\ z_j \in \{0, 1\}$ for $j \in N\}$ where $T \subseteq N$ is the support of z^T.*

(i) *If $\eta \geq 1$, none of the valid inequalities for $\mathscr{P}_{\geq}$ of the form $\sum_{j \in S} y_j \geq 1$ is violated.*

(ii) *If $\eta < 1$, the valid inequality $\sum_{j \in T} y_j \geq 1$ is violated by (x^*, y^*).*

A successful heuristic algorithm that uses this result as the basis for the generation of stronger violated inequalities is described in Crowder et al.

If we are dealing with the polytope $\mathscr{P}_{\leq}$, Proposition 11 shows how to obtain surrogate knapsack inequalities for this polytope. Violated inequalities can now be found by a two step procedure:

(i) Solve the knapsack problem of Proposition 13 to get η;

(ii) Solve the parametric knapsack problem of Proposition 11 to get t.

Now we consider the problem of generating violated flow cover facets for $\mathscr{P}_{\leq}$. First, we consider flow cover facets with $L = \phi$, and we define a family of knapsack problems, parameterized by λ. Let $\tau_j(\lambda) = m_j - x_j^* - (m_j - \lambda)^+ (1 - y_j^*)$, and

$$(P_\lambda) \qquad \begin{aligned} \xi_\lambda = \min\ & \textstyle\sum_{j=1}^n \tau_j(\lambda) z_j - \lambda \\ \text{subject to}\ & \textstyle\sum_{j=1}^n m_j z_j = d + \lambda, \quad z_j \in \{0, 1\} \quad j \in N. \end{aligned}$$

PROPOSITION 14. *Let* (ξ_λ, z^T) *be an optimal solution to the knapsack problem* (P_λ).

(i) *If* $\xi_\lambda \geq 0$, *none of the* (S, ϕ) *flow cover facets with* $\sum_{j\in S} m_j = d + \lambda$ *is violated.*

(ii) *If* $\xi_\lambda < 0$, *the* (T, ϕ) *flow cover facet is violated by* (x^*, y^*).

Proof. If z^S is a feasible solution to the knapsack problem, $\sum_{j\in S} m_j = d + \lambda$ and the objective value $\sum_{j=1}^n \tau_j(\lambda) z_j^S - \lambda = \sum_{j\in S} m_j - \sum_{j\in S} x_j^* - \sum_{j\in S} (m_j - \lambda)^+ (1 - y_j^*) - \lambda = d - \sum_{j\in S} x_j^* - \sum_{j\in S} (m_j - \lambda)^+ (1 - y_j^*) < 0$ if and only if (x^*, y^*) violates the (S, ϕ) flow cover facet.

A variety of conditions for a facet violation follow immediately as corollaries. For example, let $R = \{j \in N \mid x_j^* = m_j y_j^*\}$, and

$$\zeta_R = \min \sum_{j\in R} (1 - y_j^*) z_j$$

$$\text{subject to} \quad \sum_{j\in R} m_j z_j > d, \quad z_j \in \{0, 1\}\ j \in R.$$

COROLLARY 15. *A sufficient condition for a violated flow cover facet* (S, ϕ) *is* $\zeta_R < 1$.

Proof. If $\zeta_R < 1$, there exists an optimal solution z^S with S a minimal cover. Since $S \subseteq R$, $\tau_j(\lambda) = \lambda(1 - y_j^*)$ for all $j \in S$, and hence S is a feasible solution to (P_λ) with objective value $\lambda \zeta_R - \lambda = \lambda(\zeta_R - 1) < 0$.

Alternatively, let

$$\zeta_\lambda = \min \sum_{j\in N_\lambda} (1 - y_j^*) z_j$$

$$\text{subject to} \quad \sum_{j\in N_\lambda} m_j z_j \geq d + \lambda \cdot z_j \in \{0, 1\}\ j \in N_\lambda$$

where $N_\lambda = \{j \mid m_j > \lambda\}$.

COROLLARY 16. *A necessary condition for a violated flow cover facet* (S, ϕ) *with* $\sum_{j\in S} m_j \geq d + \lambda$ *is* $\zeta_\lambda < 1$.

Proof. If z is feasible in (P_λ), $\sum_{j=1}^n \tau_j(\lambda) z_j - \lambda \geq \sum_{j\in N_\lambda} \tau_j(\lambda) z_j - \lambda \geq \sum_{j\in N_\lambda} \lambda(1 - y_j^*) z_j - \lambda \geq \lambda(\zeta_\lambda - 1)$.

For a given choice of a cover S, the second problem is to find the (S, L) flow cover facet most violated (if at all) by (x^*, y^*). This result however, is immediate.

PROPOSITION 17. *Given a cover* S, *the most violated flow cover inequality is* (S, L^-), *where* $L^- = \{j \in N \backslash S \mid (\bar{m}_j - \lambda) y_j^* < x_j^*\}$.

The procedure for finding violated flow cover inequalities for $\mathscr{P}_\geq$ follows immediately from Proposition 9.

5. THE CASE WHERE $m_j = m$ FOR ALL $j = 1, \cdots, n$

We show in the Corollary to Theorem 5 that $\mathscr{P}_=$ has an exponential number of flow cover facets even when $m_j = m$ for all j. In this section, we show that these facets suffice to characterize $\mathscr{P}_=$.

THEOREM 18. *Suppose $d = km + r$ with $0 \le r < m$. The system*

$$(Q_=)\qquad \begin{array}{lll} \sum_{j=1}^n x_j & = d & \\ \sum_{j\in C} x_j - r\sum_{j\in C} y_j & \le k(m-r) & C \subseteq N \quad \text{with} \quad |C| \ge k+1 \\ x_j - my_j & \le 0 & j \in N \\ y_j & \le 1 & j \in N \\ x_j, y_j & \ge 0 & j \in N \end{array}$$

completely describes $\mathscr{P}_=$.

Note that we are using the (S, L) flow cover facets with $C = S \cup L$, $|S| = k + 1$ and $\lambda = m - r$.

We shall prove the result by studying the mixed 0-1 problem

$$(R_=)\qquad \max\{cx + fy \mid (x, y) \in X_=\}.$$

Without loss of generality, we suppose $c_1 \ge c_2 \cdots \ge c_n$. The following results are easy to show.

PROPOSITION 19. *The optimal solution to $(R_=)$ is of the following form:*

$$x_i = m, \quad i \in T, \quad x_{j^*} = r, \quad x_j = 0, \quad j \in N\backslash(T \cup \{j^*\})$$

$$y_j = 1 \quad \text{if} \quad x_j > 0, \quad \text{or if} \quad x_j = 0 \quad \text{and} \quad f_j > 0$$

$$y_j = 0 \quad \text{if} \quad x_j = 0 \quad \text{and} \quad f_j < 0$$

$$y_j = 0 \quad \text{or} \quad 1 \quad \text{if} \quad x_j = 0 \quad \text{and} \quad f_j = 0,$$

where $|T| = k$, and $i < j^*$ for all $i \in T$. We let F denote $\{j: x_j = 0$ and $y_j = 1\}$.

It follows that the optimal solution to $(R_=)$ can be found from the following assignment problem:

$$(A\mathscr{P}_=)\qquad \begin{array}{lll} \max \sum_{j=1}^n (mc_j + f_j)\xi_{1j} + \sum_{j=1}^n (rc_j + f_j)\xi_{2j} + \sum_{j=1}^n f_j\xi_{3j} & & \\ \text{subject to} \quad \sum_{j=1}^n \xi_{1j} & & = k \\ \sum_{j=1}^n \xi_{2j} & & = 1 \\ \xi_{1j} + \xi_{2j} + \xi_{3j} & \le 1, & j \in N \\ \xi_{ij} \ge 0, & i = 1, 2, 3, & j \in N. \end{array}$$

PROPOSITION 20. *Let* $(\xi^*_{1j}, \xi^*_{2j}, \xi^*_{3j})$ *be an integer optimal solution to* $(A\mathscr{P}_=)$ *with* $\xi^*_{1j} = 1, j \in T$, $\xi^*_{2j^*} = 1$, $\xi^*_{3j} = 1, j \in F$. *Then* (T, j^*, F) *is an optimal solution to* $(R_=)$.

Now consider the dual of $(A\mathscr{P}_=)$ with associated dual variables $(u_1, u_2, z_1, \cdots, z_n)$, namely the problem

$$(DA\mathscr{P}_=) \qquad \begin{array}{lll} \min ku_1 + u_2 + \sum_{j=1}^n z_j & & \\ u_1 \qquad\qquad + z_j \geq mc_j + f_j, & & j \in N \\ \qquad\quad u_2 \qquad + z_j \geq rc_j + f_j, & & j \in N \\ z_j \geq f_j, & & j \in N \\ z_j \geq 0, & & j \in N \end{array}$$

PROPOSITION 21. $(DA\mathscr{P}_=)$ *has an optimal dual solution in which* $u_1 + z_j = mc_j + f_j$ *for some* $j > k$.

Proof. Consider the set $T = \{j \mid \xi^*_{1j} = 1\} = \{j_1, j_2, \cdots, j_k\}$ with $j_1 < j_2 < \cdots < j_k$. If $j_k > k$, the result follows immediately from complementary slackness since $u_1 + z_j = mc_j + f_j$ whenever $\xi^*_{1j} = 1$. Therefore, we suppose that $T = \{1, 2, \cdots, k\}$, and let $\varepsilon = \min_{i>k}[u_1 + z_i - (mc_i + f_i)] = u_1 + z_t - (mc_t + f_t) \geq 0$.

Now let $u_1' = u_1 - \varepsilon$, $z_i' = z_i + \varepsilon$, $i = 1, \cdots, k$, $u_2' = u_2$, $z_i' = z_i$, $i > k$. (u_1', u_2', z') is dual feasible, and has the same objective value, and $u_1' + z_t' = mc_t + f_t$ for some $t > k$.

Now we turn to a proof of Theorem 18.

Proof of Theorem 18. We consider the linear program

$$\max\{\textstyle\sum_{j=1}^n c_j x_j + \sum_{j=1}^n f_j y_j : (x, y) \in Q_=\}$$

and show that it has an optimal solution in integers for all c and f. As before, we suppose that $c_1 \geq c_2 \geq \cdots \geq c_n$. For this ordering, it is sufficient to consider the linear program:

$$\begin{array}{lll} & \max \sum_{j=1}^n c_j x_j + \sum_{j=1}^n f_j y_j & \\ \text{subject to} & \sum_{j=1}^n x_j \qquad\qquad = d & \\ & \sum_{j=1}^p x_j - r \sum_{j=1}^p y_j \leq k(m-r) & p = k+1, \cdots, n \\ & x_j - m_y \leq 0 & j = 1, \cdots, n \\ & y_j \leq 1 & j = 1, \cdots, n \\ & x_j, y_j \geq 0 & j = 1, \cdots, n \end{array}$$

with dual variables $(v_0, v_{k+1}, \cdots, v_n, w_1, \cdots, w_n, \zeta_1, \cdots \zeta_n)$.

We define the following indices, using an optimal dual solution to $(DA\mathscr{P}_=)$ satisfying the condition of Proposition 21:

$t^* = \max\{j \mid u_1 - u_2 \le (m - r)c_j\}$

j^* is such that $\xi^*_{2j^*} = 1$

$s = \min\{j \mid c_j < u_2/r\}$, and we set $s = n + 1$ if $c_j \ge u_2/r$ for all $j \in N$.

Claim 1. $k < t^* \le j^* < s$.

To see that $k < t^*$, note that $u_1 - u_2 \le (m - r)c_j$ whenever $u_1 + z_j = mc_j + f_j$, and $u_2 + z_j \ge rc_j + f_j$. By Proposition 21, there exists a $t > k$ for which this inequality holds, and hence $t^* \ge t > k$.

To see that $t^* \le j^*$, note that $\xi_{2j^*} = 1$ implies that $u_2 + z_j{}^* = rc_{j^*} + f_{j^*}$, and dual feasibility implies that $u_1 + z_{j^*} \ge mc_{j^*} + f_{j^*}$, so that $u_1 - u_2 \ge (m - r)c_{j^*}$. If strict inequality holds, $t^* < j^*$ because of the ordering of the c_j, and otherwise we can reorder the variables so that $t^* = j^*$.

Finally, note that $u_2 + z_{j^*} = rc_{j^*} + f_{j^*}$, and $z_{j^*} \ge f_{j^*}$, so that $u_2 \le rc_{j^*}$ and hence $j^* < s$.

Now we define a dual solution for our linear program:

$$
\begin{aligned}
v_0 &= u_2/r \\
v_p &= 0, \quad p = k + 1, \cdots, t^* - 1 \\
v_{t^*} &= [(u_1 - u_2)/(m - r)] - c_{t^*+1} \\
v_p &= c_p - c_{p+1}, \quad t^* < p < s - 1 \\
v_{s-1} &= c_{s-1} - u_2/r \\
v_p &= 0, \quad p \ge s \\
w_j &= (c_j - (u_1 - u_2)/(m - r))^+, \quad j = 1, \cdots, n \\
\zeta_j &= z_j, \quad j = 1, \cdots, n.
\end{aligned}
$$

Claim 2

This solution is dual feasible, so that

$$v_0 + \textstyle\sum_{p=\max\{j,k+1\}}^{n} v_p + w_j \ge c_j, \quad j = 1, \cdots, n \tag{1}$$

$$-r \textstyle\sum_{p=\max\{j,k+1\}}^{n} v_p - mw_j + \zeta_j \ge f_j, \quad j = 1, \cdots, n \tag{2}$$

$$v_{k+1}, \cdots, v_n, w, \zeta \ge 0. \tag{3}$$

Note first that the nonnegativity constraints (3) are satisfied.

For $j = 1, \cdots, t^*$, $\sum_{p=\max\{j,k+1\}}^{n} v_p = ([(u_1 - u_2)/(m - r)] - c_{t^*+1}) + (c_{t^*+1} - c_{t^*+2}) + \cdots + (c_{s-1} - u_2/r) = [(u_1 - u_2)/(m - r)] - u_2/r$.

For $j = t^* + 1, \cdots, s - 1$, $\sum_{p=\max\{j,k+1\}}^{n} v_p = c_j - u_2/r$, and for $j = s, \cdots, n$, $\sum_{p=\max\{j,k+1\}}^{n} v_p = 0$.

Now to verify (1) and (2) for $j = 1, \cdots, t^*$, we have

$$\frac{u_2}{r} + \left(\frac{u_1 - u_2}{m - r} - \frac{u_2}{r}\right) + \left(c_j - \frac{u_1 - u_2}{m - r}\right) = c_j$$

and

$$-r\left(\frac{u_1 - u_2}{m - r} - \frac{u_2}{r}\right) - m\left(c_j - \frac{u_1 - u_2}{m - r}\right) + z_j$$
$$= (u_1 - u_2) + u_2 - mc_j + z_j = u_1 + z_j - mc_j \geq f_j.$$

To verify (1) and (2) for $j = t^* + 1, \cdots, s - 1$, we have:

$$u_2/r + (c_j - u_2/r) + 0 = c_j$$

and $-r(c_j - u_2/r) - 0 + z_j = u_2 + z_j - rc_j \geq f_j$.

To verify (1) and (2) for $j = s, \cdots, n$, we have:

$$u_2/r + 0 + 0 \geq c_j$$

and $-0 + 0 + z_j \geq f_j$.

Finally, we note that the objective value of our dual feasible solution is $dv_0 + \sum_{p=k+1}^{n} k(m - r)v_p + \sum_{j=1}^{n} \zeta_j = (km + r)u_2/r + k(m - r) \cdot ([(u_1 - u_2)/(m - r)] - (u_2/r)) + \sum_{j=1}^{n} z_j = ku_1 + u_2 + \sum_{j=1}^{n} z_j$, which is the value of the optimal solution to $(DA\mathscr{P}_=)$, and hence of the primal integer solution given in Proposition 19.

6. EXTENSIONS AND AN EXAMPLE

Given a general fixed cost arc network problem on a network $G = (V, E)$ where $V = S \cup R \cup I$ with sources S, sinks R and intermediate nodes I, we are interested in the problem:

$$\begin{array}{lll} \min \sum_{e \in E} c_e x_e & + \sum_{e \in E} f_e y_e & \\ \text{subject to } \sum_{V_i} x_e & - \sum_{V_i'} x_e & = b_i \quad i \in V \\ & 0 \leq x_e \leq m_e y_e, \quad y_e \in \{0, 1\}. & e \in E \end{array}$$

where $V_i = \{e \in E \mid e = (i, j) \text{ for some } j \in V\}$, $V_i' = \{e \in E \mid e = (k, i) \text{ for some } k \in V\}$, $b_i > 0$ for $i \in S$, $b_i < 0$ for $i \in R$, $b_i = 0$ for $i \in I$ and $\sum_{i \in V} b_i = 0$.

For every $U \subseteq V$, one obtains the relaxation

$$\sum_{e \in (U, \bar{U})} x_e - \sum_{e \in (\bar{U}, U)} x_e = \sum_{i \in U \cap R} |b_i| - \sum_{i \in U \cap S} b_i,$$

and, in particular,

$$\sum_{e \in (U, \bar{U})} x_e \geq \sum_{i \in U \cap R} |b_i| - \sum_{i \in U \cap S} b_i$$
$$0 \leq x_e \leq m_e y_e, \quad y_e \in \{0, 1\}$$

which is clearly an example of $X_{\geq}$.

Hence, for every $U \subseteq V$, we can generate valid inequalities for this problem. When $|U| = 1$ or $|V| - 1$, we obtain the case of a single node discussed in the introduction.

The problem of generating appropriate choices of U is still being examined; however, note that if $\mathscr{U} = \{U \subseteq V \mid \sum_{i \in U \cap R} |b_i| - \sum_{i \in U \cap S} b_i > 0\}$ one immediately obtains the valid inequalities

$$\sum_{e \in (U, \bar{U})} y_e \geq 1 \quad \text{for all} \quad U \in \mathscr{U}.$$

The violation of such inequalities can be tested by max flow calculations, suggesting a tie-in with the work on the tree polytope reported in Padberg and Wolsey.

It is also possible to extend the facets generated in this paper by lifting (Padberg [1979], Wolsey, and Zemel) to obtain facets for problems such as

$$\sum_{V_i} x_e - \sum_{V'_i} x_e = b_i$$

$$0 \leq x_e \leq m_e y_e, \quad y_e \in \{0, 1\}.$$

These extensions are at present under investigation. Some preliminary computational testing based on the generation of violated "surrogate knapsack" and "flow cover" inequalities is also under way.

Example

Consider the problem with data

$$n = 5, \quad d = 10, \quad m_1 = m_2 = 5, \quad m_3 = 4, \quad m_4 = 3, \quad m_5 = 1.$$

(a) The inequality

$$y_1 + y_2 + y_4 + y_5 \geq 2$$

defines a facet of the surrogate knapsack polytope: $\text{conv}\{y \in R^5 \mid 5y_1 + 5y_2 + 4y_3 + 3y_4 + 1y_5 \geq 10, y_j \in \{0, 1\} \text{ for } j \in N\}$, but $y_1 + y_2 + y_4 + y_5 = 2$ implies $x_5 = y_5$ for all $(x, y) \in X_{\geq}$, so it is a valid inequality for $\mathscr{P}_{\geq}$, but not a facet (Theorem 2).

(b) The inequality

$$x_1 + x_2 + x_3 + x_4 - 3y_1 - 3y_2 - 2y_3 - 1y_4 \leq 4$$

is a facet of $\mathscr{P}_{=}$, which can be established by taking $S = \{1, 3, 4\}$ and $L = \{2\}$ in Theorem 5.

(c) It is a facet of $\mathscr{P}_{\leq}$ by Proposition 11.

(d) It becomes the facet

$$x_5 + 3y_1 + 3y_2 + 2y_3 + 1y_4 \geq 6$$

of $\mathscr{P}_{\geq}$ by Proposition 10.

(e) Given the point $(x, y) = (3, 0, 4, 3, 0; 0.6, 0, 1, 1, 0)$ lying in the linear

programming relaxation of $X_=$, the problem

$$\eta = \min\{0.6z_1 + 0z_2 + z_3 + z_4 + 0z_5 \mid 5z_1 + 5z_2 + 4z_3 + 3z_4 + 1z_5 > 8,\ z_i \in \{0, 1\}\}$$

has optimal solution $S = \{1, 2\}$ with $\eta = 0.6$. Hence, the valid inequality $y_1 + y_2 \geq 1$ is violated, using Proposition 13.

(f) Given the point $(x, y) = (3.75, 0.25, 4, 1.5, 0.5; 0.75, 0.25, 1, 0.5, 0.5)$, the problem

$$\zeta_1 = \min\{0.25z_1 + 0.75z_2 + 0z_3 + 0.5z_4 + 0.5z_5 \mid 5z_1 + 5z_2 + 4z_3 + 3z_4 + 1z_5 \geq 11,\ z_i \in \{0, 1\}\}$$

has optimal solution $S = \{1, 3, 4\}$ with $\zeta_1 = 0.75$. After checking, we see that the flow cover facet with $S = \{1, 3, 4\}$

$$x_1 + x_3 + x_4 + 3(1 - y_1) + 2(1 - y_3) + (1 - y_4) \leq 10$$

is violated, using Corollary 16.

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